

NAME: Ayilara Busari Dare

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Student ID: IDEAS/24/28133

Lab 3: Network Security Protocols and Configuration

Network security protocols are defined as sets of rules and procedures designed to ensure the confidentiality, integrity, and authenticity of data transmitted over networks. These protocols protect network traffic by using various mechanisms to secure communication, thereby preventing unauthorized access and ensuring that data remains intact and trustworthy as it moves through potentially vulnerable environments.

The importance of each of these following protocols in securing communications

- **SSL/TLS: Securing data in transit.** SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are critical encryption protocols that secure and authenticate communications over the internet. They protect sensitive information, such as personal data and payment details, during online transactions by encrypting the data transmitted between users and servers. This ensures that e-commerce sites, online banking services, and other platforms can operate safely, safeguarding user privacy and preventing data breaches.
- **IPsec: Protecting IP packets.** IPsec is important because it provides a suite of protocols designed to secure data communications over networks, especially when transmitting sensitive information across public networks like the Internet. It offers functionalities such as encryption, data integrity, and authentication, making it essential for applications like Virtual Private Networks (VPNs) and securing financial transactions, medical records, and corporate communications.
- **SSH: Secure shell for secure remote access.** SSH (Secure Shell) is crucial for establishing secure connections over potentially insecure networks. It provides encrypted communication to protect sensitive data from security breaches, making it essential in environments like cloud computing and server administration, where secure remote access and management are necessary.
- **VPN: Virtual Private Network for secure connections.** A VPN (Virtual Private Network) is important for several reasons: it secures your internet connection, protects your personal data from hackers and snoopers, hides your IP address for enhanced privacy, and allows access to geo-restricted content. Additionally, it can help prevent data throttling by your Internet Service Provider (ISP) and enhance your online anonymity.

- **802.1X: Port-based network access control**, 802.1X is a network authentication protocol that enhances security by requiring unique credentials for each device attempting to access a network. It provides stronger access control and is more secure than traditional methods such as basic Wi-Fi authentication, as it prevents unauthorized access and protects the network from potential hackers.

2. Configuring SSL/TLS on a Web Server

Apache local web server was set up on a Linux virtual machine, here are the steps used to achieve secure websites using SSL/TLS and redirect http to https traffic.

Apache2 was installed using **Sudo apt install apache2 -y** and open a web browser type my ip address which is 10.0.2.15, then not secure apache web server page appear.

Configuration of the web server to support HTTPS

SSL was enable using openssl for the configuration as show in the below picture

```
Engr.ABD@kali: ~  
File Actions Edit View Help  
  
(Engr.ABD@kali)-[~]  
$ sudo apt update && sudo apt upgrade -y  
  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
All packages are up to date.  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
  
(Engr.ABD@kali)-[~]  
$ sudo apt install apache2 -y  
  
apache2 is already the newest version (2.4.62-3).  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
  
(Engr.ABD@kali)-[~]  
$ sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/ssl/private/selfsigned.key -out /etc/ssl/certs/selfsigned.crt  
  
Debian's Apache2 default configuration is different from the upstream default configuration. The difference is usually minor, but it may cause some problems if you are not familiar with the configuration files. You can find more information about this by accessing the documentation at https://httpd.apache.org/docs/2.4/.
```

Figure 1: installation of apache2 and ssl using openssl for configuration.

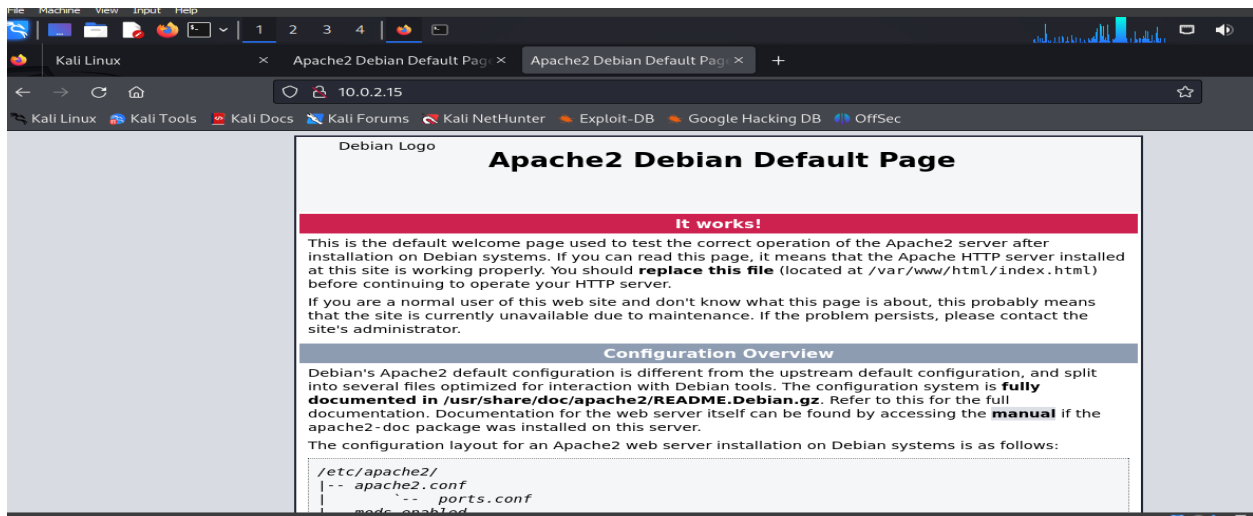
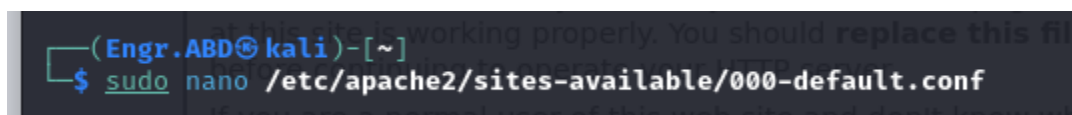
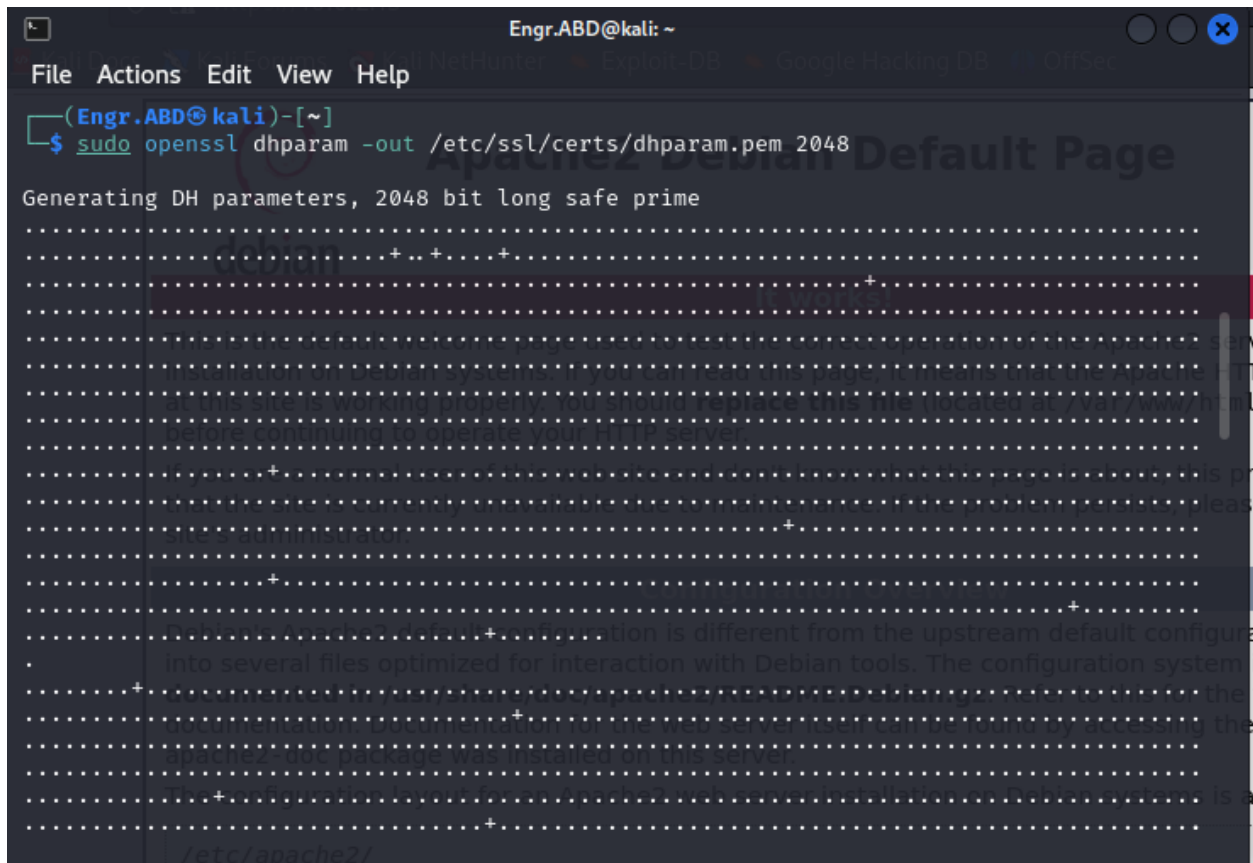
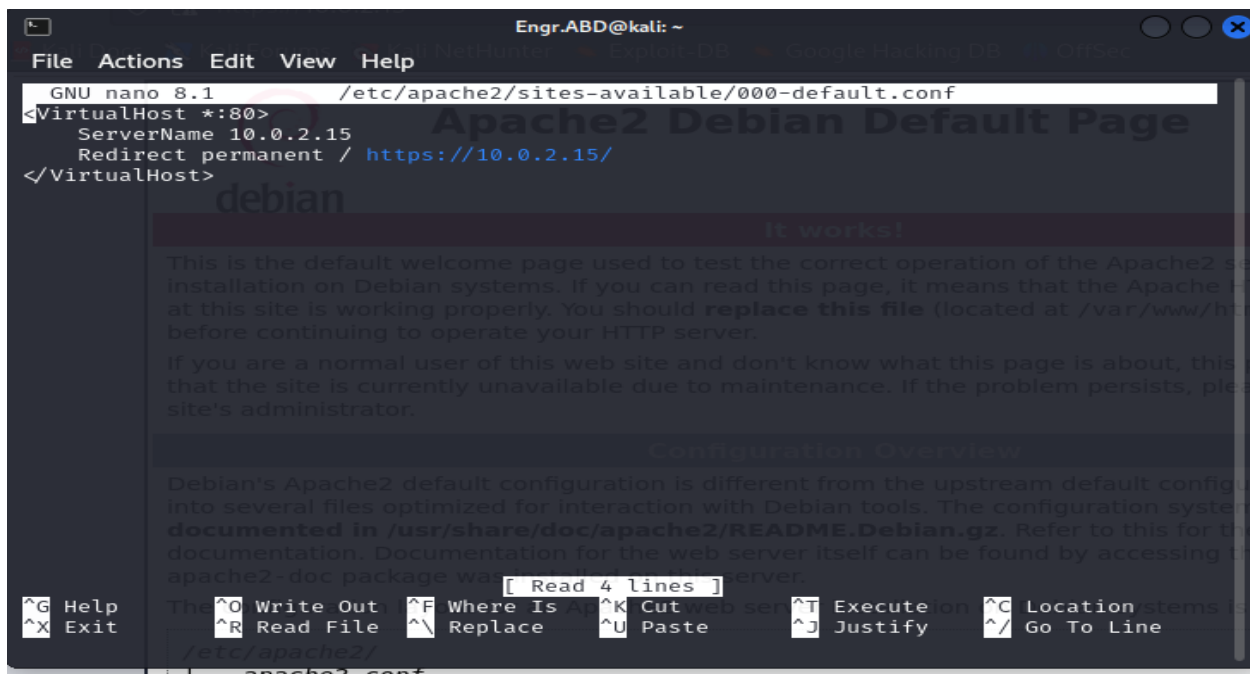


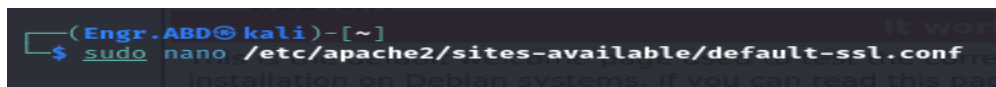
Figure 2: Apache2 web server page not secure



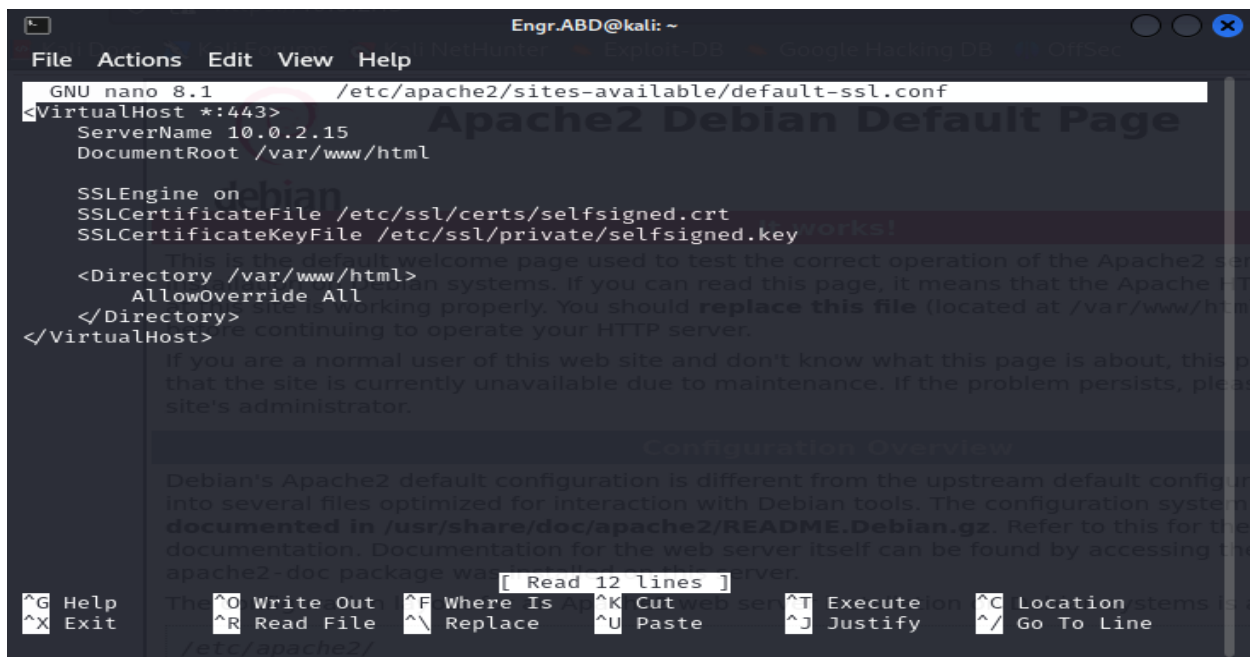


```
Engr.ABD@kali: ~
File Actions Edit View Help
GNU nano 8.1 /etc/apache2/sites-available/000-default.conf
<VirtualHost *:80>
    ServerName 10.0.2.15
    Redirect permanent / https://10.0.2.15/
</VirtualHost>
```

Figure 3: host to listen to port 80 and server name to 10.0.2.15



```
(Engr.ABD@kali)-[~]
$ sudo nano /etc/apache2/sites-available/default-ssl.conf
```



```
Engr.ABD@kali: ~
File Actions Edit View Help
GNU nano 8.1 /etc/apache2/sites-available/default-ssl.conf
<VirtualHost *:443>
    ServerName 10.0.2.15
    DocumentRoot /var/www/html

    SSLEngine on
    SSLCertificateFile /etc/ssl/certs/selfsigned.crt
    SSLCertificateKeyFile /etc/ssl/private/selfsigned.key

    <Directory /var/www/html>
        AllowOverride All
    </Directory>
</VirtualHost>
```

Figure 4: configure host to listen to port 443

```
(Engr.ABD@kali)-[~]
$ sudo netstat -tuln | grep ':80\|:443'

tcp6      0      0 :::443          :::*             LISTEN
tcp6      0      0 :::80           :::*             LISTEN
```

Figure 5: this command use to see the port that is listen

```
(Engr.ABD@kali)-[~]
$ sudo a2ensite 000-default.conf
sudo a2ensite default-ssl.conf

Site 000-default already enabled
Enabling site default-ssl:2 default configuration is different from
To activate the new configuration, you need to run:
systemctl reload apache2

(Engr.ABD@kali)-[~]
$ sudo systemctl restart apache2
```

Figure 6: Enable and activate the new configuration and restart apache2

This can also be carryout using Lets Encrypt, using certbot but you need a domain name before one can use lets encrypt. You can check for your public IP address to register for domain. As shown below.

```
Engr.ABD@kali: ~
File Actions Edit View Help
Engr.ABD@kali: ~ x Engr.ABD@kali: ~ x

(Engr.ABD@kali)-[~]
$ curl ifconfig.me

102.89.69.172

(Engr.ABD@kali)-[~]
$
```

Figure 7: how to check for public ip address

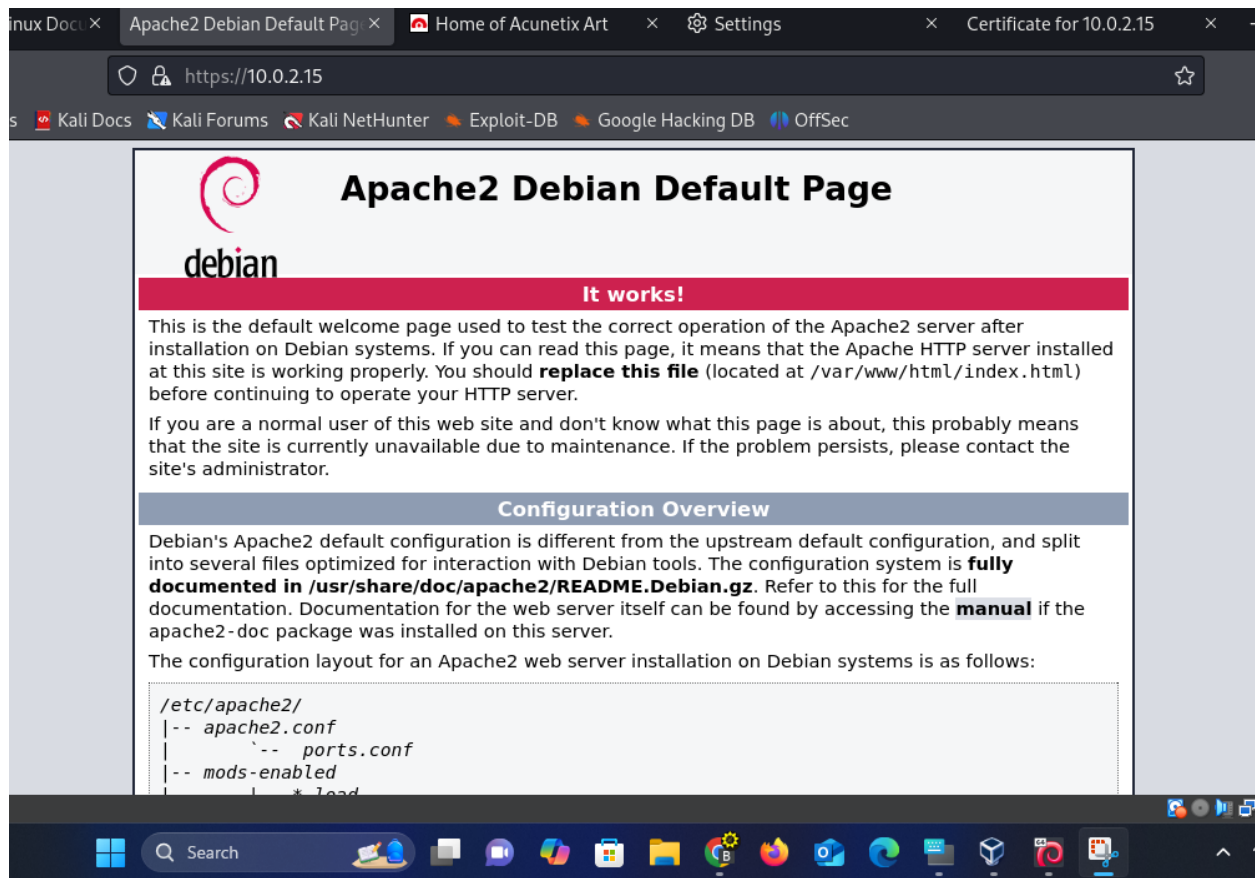


Figure 8: Redirect HTTP to HTTPS for web server page secure

Subject Name	
Country	NG
State/Province	kwara
Locality	ilorin
Organization	ICFDA
Organizational Unit	cyber
Common Name	10.0.2.15
Email Address	aminatayilara@gmail.com
Issuer Name	
Country	NG
State/Province	kwara
Locality	ilorin
Organization	ICFDA
Organizational Unit	cyber
Common Name	10.0.2.15
Email Address	aminatayilara@gmail.com
Validity	
Not Before	Sat, 14 Dec 2024 21:34:58 GMT
Not After	Sun, 14 Dec 2025 21:34:58 GMT
Public Key Info	
Algorithm	RSA
Key Size	2048
Exponent	65537
Modulus	93:CB:DD:ED:DF:3A:47:3C:21:25:2B:A2:6E:63:41:3C:F9:4B:B6:11:BE:8F:2B:C3...
Miscellaneous	
Serial Number	7A:F7:B6:EE:3C:F5:28:64:D8:76:13:3F:4A:56:18:22:76:8E:3A:A5
Signature Algorithm	SHA-256 with RSA Encryption
Version	3
Download	PEM (.cert) PEM (.chain)

Firefox

about:certificate?cert=MIID9zCCAt%2BgAwIBAgIUeve27jz1KGTydhM%2FSlYYInaOOqUwDQYJKoZIhvi...

80%

Kali Docs

Kali Forums

Kali NetHunter

Exploit-DB

Google Hacking DB

OffSec

Serial Number	7A:F7:B6:EE:3C:F5:28:64:D8:76:13:3F:4A:56:18:22:76:8E:3A:A5
Signature Algorithm	SHA-256 with RSA Encryption
Version	3
Download	PEM (.cert) PEM (.chain)
Fingerprints	
SHA-256	4A:90:4C:4B:F0:5F:B2:69:5D:F5:B3:7A:A5:47:38:9C:BA:E7:2D:63:83:13:1B:EE...
SHA-1	65:84:60:83:6F:60:DA:69:BF:CA:C4:2C:3D:C7:A3:98:EB:14:9E:20
Basic Constraints	
Certificate Authority	Yes
Subject Key ID	
Key ID	DF:F3:4C:A2:3A:20:49:BC:72:5B:71:A2:9E:5C:41:D3:DA:E7:B6:DA
Authority Key ID	
Key ID	DF:F3:4C:A2:3A:20:49:BC:72:5B:71:A2:9E:5C:41:D3:DA:E7:B6:DA

Figure 9: certificate

Configuring SSH for Secure Remote Access

The ssh was install on kali linux using **sudo apt install openssh@server**, after installation, the ssh was start using command **sudo systemctl start ssh** or **sudo service ssh start** depending on the one you wish to use. Then the status on the ssh was check the verify it activeness using **command sudo systemctl status ssh**.

```
(hackb@kali)-[~]
$ sudo service ssh start

(hackb@kali)-[~]
$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2024-12-20 09:12:13 EST; 1min 1s ago
     Invocation: 25f076519a22435cbc06812b3715576f
       Docs: man:sshd(8)
            man:sshd_config(5)
    Process: 9837 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 9839 (sshd)
      Tasks: 1 (limit: 2222)
     Memory: 1.4M (peak: 2.3M)
        CPU: 158ms
    CGroup: /system.slice/ssh.service
            └─9839 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 20 09:12:13 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell:
Dec 20 09:12:13 kali sshd[9839]: Server listening on 0.0.0.0 port 22.
Dec 20 09:12:13 kali sshd[9839]: Server listening on :: port 22.
Dec 20 09:12:13 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell:
```

Figure 10. installation of ssh and systemctl

Adjust settings to enhance security: the port was Change from the default port (22) to a non-standard port 2222, Disable root login by set it to NO and Implement public key authentication instead of password-based authentication.

```

(hackb@kali)-[~]
$ cd /etc/ssh

(hackb@kali)-[/etc/ssh]
$ ls
moduli          sshd_config      ssh_host_ed25519_key
moduli.dpkg-old sshd_config.d    ssh_host_ed25519_key.pub
ssh_config      ssh_host_ecdsa_key  ssh_host_rsa_key
ssh_config.d    ssh_host_ecdsa_key.pub  ssh_host_rsa_key.pub

(hackb@kali)-[/etc/ssh]
$ sudo nano ssh_config

(hackb@kali)-[/etc/ssh]
$ sudo nano sshd_config

(hackb@kali)-[/etc/ssh]
$ sudo nano sshd_config

```

```

File  Actions  Edit  View  Help
GNU nano 8.1      sshd_config
# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

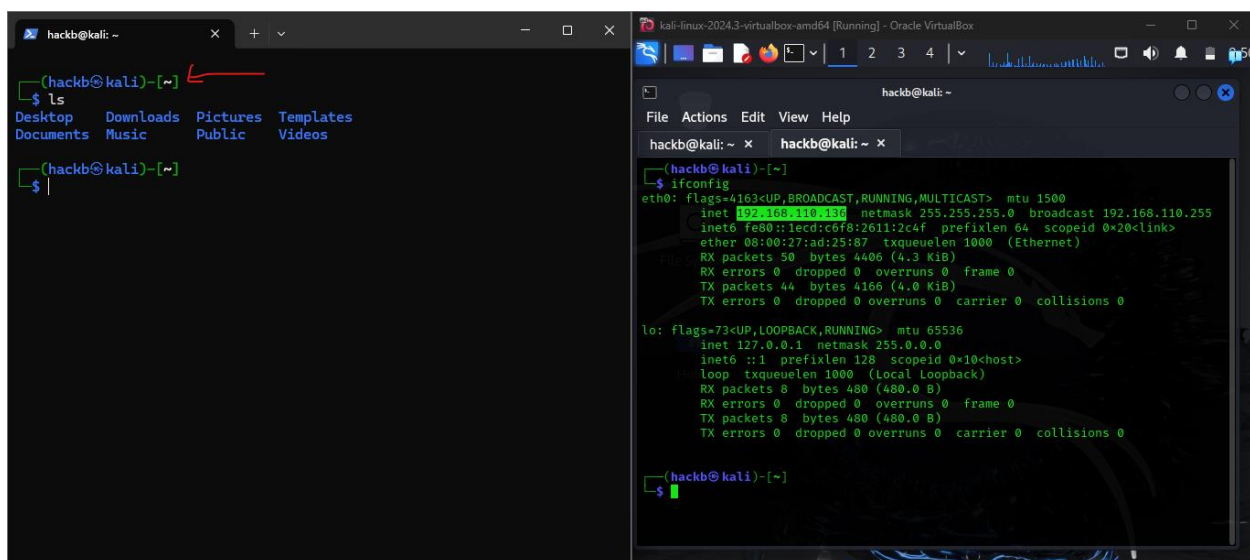
#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
[ Read 122 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut
^X Exit      ^R Read File  ^\ Replace    ^U Paste
^T Execute
^J Justify

```

```
(hackb@kali)-[/etc/ssh]
$ sudo systemctl reload ssh

(hackb@kali)-[/etc/ssh]
$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/hackb/.ssh/id_rsa):
Created directory '/home/hackb/.ssh'.
Enter passphrase for "/home/hackb/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/hackb/.ssh/id_rsa
Your public key has been saved in /home/hackb/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:Wc1lI30KJhYfx8ccJYh2q3NzNu7e0nNoil5hSaJMvtY hackb@kali
The key's randomart image is:
+--[RSA 3072]-----+
|      .o++Bo.      |
|      oo=+0+o      |
|      o.=.*o.      |
|      .*.+o.      |
|      S. +         |
|      ooo.+        |
|      ooE=.o.      |
|      .oo.=.      |
|      .oo++o.      |
+--[SHA256]-----+
```



Using 802.1X for Network Access Control

Simulated a network with a switch that supports 802.1X was carried out on packet tracker. The server configures to AAA RADIUS server to handle authentication requests. client devices was set to the network and verification was done using terminal, it shows that only authenticated devices can access resources using username and password.

OK bytes of WebUI ODM Files at webui:.

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int Gig0/0/0
Router(config-if)#ip ad
% Incomplete command.
Router(config-if)#ip address 200.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
```

Copy

Paste

Assign a local username And password, just in case our AAA server fails, we will be using it to access the router.


```
RW01(config)#
RW01(config)#
RW01(config)#aaa authentication?
authentication
RW01(config)#aaa authentication login ?
WORD      Named authentication list.
default   The default authentication list.
RW01(config)#aaa authentication login default ?
enable     Use enable password for authentication.
group      Use Server-group.
local      Use local username authentication.
local-case Use case-sensitive local username authentication.
none       NO authentication.
RW01(config)#aaa authentication ?
enable     Set authentication lists for enable.
login      Set authentication lists for logins.
ppp        Set authentication lists for ppp.
RW01(config)#aaa authentication enable ?
default    The default authentication list.
RW01(config)#aaa authentication enable default ?
enable     Use enable password for authentication.
group      Use Server-group.
none       NO authentication.
RW01(config)#aaa authentication enable default group ?
radius     Use list of all Radius hosts.
tacacs+    Use list of all Tacacs+ hosts.
RW01(config)#aaa authentication enable default group radius ?
enable     Use enable password for authentication.
group      Use Server-group.
local      Use local username authentication.
local-case Use case-sensitive local username authentication.
none       NO authentication.
<cr>
RW01(config)#aaa authentication enable default group radius local
RW01(config)#
```

Copy

Paste

```

-----
RW01(config)#ip domain name
% Incomplete command.
RW01(config)#ip domain name ayilara.com
RW01(config)#ssh version 2
^
% Invalid input detected at '^' marker.

RW01(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
RW01(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
RW01(config)#crypto key generate rsa
The name for the keys will be: RW01.ayilara.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

RW01(config)#
*Mar 1 1:46:57.990: %SSH-5-ENABLED: SSH 2 has been enabled
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)#

```

```

-----
RW01(config)#radius-server host 200.0.0.2 key 123
^

```

The above radius server, tell the router where to look for a radius server, don't forget, it has to be the same for the router and AAA server.

```
-----
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)#
RW01(config)# vty 0 4
      ^
% Invalid input detected at '^' marker.

RW01(config)#line vty 0 4
RW01(config-line)#transport input ssh
RW01(config-line)#login authentication default
  AAA: Warning authentication list default is not defined for LOGIN
RW01(config-line)#login ?
  authentication  authenticate using aaa method list
  local          Local password checking
  <cr>
RW01(config-line)#login ?
  authentication  authenticate using aaa method list
  local          Local password checking
  <cr>
RW01(config-line)#login ?
  authentication  authenticate using aaa method list
  local          Local password checking
  <cr>
RW01(config-line)#login authentication ?
  WORD           authenticate using aaa method list
  default        authenticate using aaa default list
RW01(config-line)#login authentication default
RW01(config-line)#
```

```
RW01(config)#ip domain name
% Incomplete command.
RW01(config)#ip domain name ayilara.com
RW01(config)#ssh version 2
      ^
% Invalid input detected at '^' marker.

RW01(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
RW01(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
RW01(config)#crypto key generate rsa
The name for the keys will be: RW01.ayilara.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

RW01(config)#
*Mar 1 1:46:57.990: %SSH-5-ENABLED: SSH 2 has been enabled
RW01(config)#
```

FastEthernet0 Connection:(default port)

```
Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::260:70FF:FEB1:E998
IPv6 Address.....: ::
IPv4 Address.....: 200.0.0.10
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        200.0.0.1
```

Bluetooth Connection:

```
Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0
```

C:\> ping 200.0.0.1

Pinging 200.0.0.1 with 32 bytes of data:

```
Reply from 200.0.0.1: bytes=32 time=25ms TTL=255
Reply from 200.0.0.1: bytes=32 time<1ms TTL=255
Reply from 200.0.0.1: bytes=32 time<1ms TTL=255
Reply from 200.0.0.1: bytes=32 time=1ms TTL=255
```

Ping statistics for 200.0.0.1:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 25ms, Average = 6ms
```

C:\>

```

[Connection to 200.0.0.1 closed by foreign host]
C:\>ssh -l aylara 200.0.0.1

Password:
RW01>
RW01>
RW01>en
Username:
Password:
% Error in authentication.
RW01>en
Username:
Password:
RW01# show run
Building configuration...

Current configuration : 940 bytes
!
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname RW01
!
!
!
!
!
!
!
aaa new-model
!
aaa authentication login default group radius local
aaa authentication enable default group radius local
!
!
RW01#

```

Lab 4: Intrusion Detection Systems (IDS) and Traffic Analysis on Linux

Install and Configure Snort

The intrusion detection system was installed **using `sudo apt-get install snort` or `sudo apt-get install snort -y`**, and the Snort was open for configuration and file for editing through **`sudo nano /etc/snort/snort.conf`** and it was configure to listing to OWASP VM IP address under HOME_NET.

The IDS is Run Snort in IDS mode to monitor the network: `sudo snort -A console -c /etc/snort/snort.conf -i eth0`